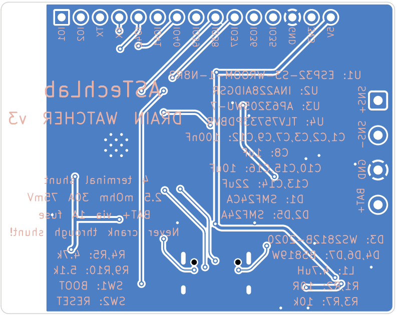
IO1
IO2
TX
RX
IO42
IO41
IO40
IO39
IO38
IO37
IO36
IO35
GND
3V3
5V
ASTechLab
DRAIN WATCHER v3
U1: ESP32-S3-WROOM-1-N8R8
U2: INA228AIDGSR
U3: AP63205WU-7
U4: TLV75733PDBVR
C1,C2,C3,C7,C9,C12: 100nF
C8: 1uF
C10,C15,C16: 10uF
C13,C14: 22uF
D1: SMF24CA
D2,D5: SMF24A
D3: WS2812B-2020
D4,D6,D7: BS819W
L1: 4.7uH
R1,R2: 10R
R3,R7: 10k
R4,R5: 4.7k
R9,R10: 5.1k
SW1: BOOT
SW2: RESET
4 terminal shunt
2.5 mOhm 30A 75mV
BAT+ via 1A fuse
Never crank through shunt!
SNS+
SNS-
GND
BAT+